

Oracle Database 12c: Block Corruption & RMAN Recovery

Technical Reference Manual for Simulating, Detecting, and Restoring Corrupted Data Blocks via DBVERIFY and RMAN

1. Baseline Setup & Full Database Backup

Before introducing corruption, verify datafiles and tables structures, then execute a full RMAN backup including archived redo logs to establish a recovery baseline.

DATAFILE STRUCTURE VERIFICATION

```
SQL> select name from v$datafile;
NAME
-----
/u01/app/oracle/oradata/sha/system01.dbf
/u01/app/oracle/oradata/sha/sysaux01.dbf
/u01/app/oracle/oradata/sha/undotbs01.dbf
/u01/app/oracle/oradata/sha/souvik01.dbf
/u01/app/oracle/oradata/sha/users01.dbf
```

SAMPLE TABLE RECORDS ('TEST1')

```
SQL> select * from test1;
EMP_NUMBER NAME          JOB              SALARY JOINING_D   DEPT_NO COMM  CITY
-----
1 Arun Sen      Manager         50000 01-JAN-83    10 .5   Bangalore
2 Sumit Roy     Manager         45000 02-MAY-86    20 .5   Mumbai
3 Soma Bose     Sales           15000 04-FEB-83    30 .4   Chennai
4 Sushil Sen    Clerk           20000 14-AUG-89    40 .4   Pune
5 ANIL Mitra    Analyst         67000 13-SEP-89    50 .4   Kolkata
6 Sunil Roy     Salesman        11000 28-NOV-90    60 .4   Ranchi
7 Sumi Pal      Mechanical      35000 02-NOV-91    70 .4   Hyderabad
8 Priti Sen     IT              35050 14-FEB-92    80 .4   Delhi
9 Aksay Hood    Analyst         22000 12-JUN-89    90 .2   Rajasthan
10 Raj Sankar   Electrical      28000 13-JUL-96    10 .2   Gujrat
11 Asha Sen     Marketing       48000 08-OCT-12    30 .3   Gujrat
```

RMAN FULL BACKUP WITH ARCHIVED LOGS

```
RMAN> backup database plus archivelog;
```

Backup Summary: Completed successfully with backup sets for archived logs (Key 95), datafiles 1, 3, 4, 5, 7 (Key 96, Tag **TAG20210714T110518**), and control file autobackup.

2. Initial Block Verification (DBVERIFY)

Running **dbv** against datafile 5 (**souvik01.dbf**) prior to any corruption indicates zero failing pages and 0 marked corrupt blocks.

DBVERIFY COMMAND & OUTPUT

```
[oracle@localhost Desktop]$ dbv file=/u01/app/oracle/oradata/sha/souvik01.dbf
```

```
DBVERIFY: Release 12.2.0.1.0 - Production on Wed Jul 14 11:06:44 2021  
Total Pages Examined          : 6400  
Total Pages Processed (Data)  : 1  
Total Pages Failing   (Data)  : 0  
Total Pages Marked Corrupt    : 0
```

3. Simulating Physical Block Corruption

We purposefully inject binary corruption into block 100 of datafile 5 using the Unix **dd** utility to simulate storage corruption.

BINARY OVERWRITE VIA DD

```
[oracle@localhost Desktop]$ . oraenv
ORACLE_SID = [oracle] ? sha
[oracle@localhost Desktop]$ dd of=/u01/app/oracle/oradata/sha/souvik01.dbf bs=8192 conv=notrunc seek=100
jhbfukjksjkfshb
jhbjdfhgjdkfh479857
<
```

Corruption Injected: Block 100 has been overwritten with invalid header signatures, s

4. Post-Corruption Verification Check

Running dbv again immediately detects the corruption on block 100.

DBVERIFY DETECTION OUTPUT

```
[oracle@localhost Desktop]$ dbv file=/u01/app/oracle/oradata/sha/souvik01.dbf

DBVERIFY - Verification starting : FILE = /u01/app/oracle/oradata/sha/souvik01.dbf
Page 100 is marked corrupt
Corrupt block relative dba: 0x01400064 (file 5, block 100)
Bad header found during dbv:
Data in bad block: type: 106 format: 0 rdba: 0x6a6b7566
...
Total Pages Marked Corrupt      : 1
```

5. RMAN Block Recovery (`blockrecover`)

Unlike full datafile recovery, RMAN block recovery allows pinpoint restoration of only the corrupted block without

RMAN BLOCK RECOVERY EXECUTION

```

[oracle@localhost Desktop]$ rman target /
RMAN> blockrecover datafile 5 block 100;

Starting recover at 14-JUL-21
using channel ORA_DISK_1
searching flashback logs for block images until SCN 10339124
finished flashback log search, restored 0 blocks

channel ORA_DISK_1: restoring block(s)
channel ORA_DISK_1: specifying block(s) to restore from backup set
restoring blocks of datafile 00005
channel ORA_DISK_1: reading from backup piece /u01/app/oracle/fast_recovery_area/sha/SHA/backupset/2021_07_14/o
channel ORA_DISK_1: restored block(s) from backup piece 1
channel ORA_DISK_1: block restore complete, elapsed time: 00:00:01

starting media recovery
media recovery complete, elapsed time: 00:00:01

Finished recover at 14-JUL-21

```

Recovery Stage	Operation Performed	Status
Flashback Log Search	Scanned flashback logs up to SCN 10339124.	0 blocks found
Block Restoration	Extracted block 100 from backup piece tag TAG20210714T110518	Restored successfully
Media Recovery	Applied archived redo logs to roll forward the block.	Media recovery complete

6. Final Block Verification Check

Re-running DBVERIFY confirms that block corruption has been completely cleared and zero pages remain marked corrupt.

FINAL DBVERIFY VALIDATION

```

[oracle@localhost Desktop]$ dbv file=/u01/app/oracle/oradata/sha/souvik01.dbf

DBVERIFY: Release 12.2.0.1.0 - Production on Wed Jul 14 11:12:53 2021
DBVERIFY - Verification starting : FILE = /u01/app/oracle/oradata/sha/souvik01.dbf
DBVERIFY - Verification complete

Total Pages Examined      : 6400
Total Pages Failing      (Data) : 0
Total Pages Marked Corrupt : 0

```

Conclusion: RMAN block recovery successfully repaired the corrupted block 100 in data